University of Florida, Department of Mathematics

Topology First-Year Exam, January 2026, Part 2

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that for every continuous map $f : S^1 \to S^1$ there is $n \in \mathbb{Z}$ such that f is homotopic to $z^n : S^1 \to S^1$.

2. Derive the Brouwer Fixed Point Theorem for B^2 from the Borsuk–Ulam Theorem for S^2.

3. Show that S^2 is not homeomorphic to S^3.

4. Show that every map $f : S^3 \to S^1 \times S^1 \times S^1$ of the 3-sphere to the 3-torus is null-homotopic.

5. Show that the Stone–Čech compactification of the reals is not metrizable.

Answer the following with complete definitions or statements or proofs.

6. Compute the fundamental group $\pi_1(S^2)$.

7. Show that any two continuous maps $f, g : X \to I^n$ to the n-cube are homotopic.

8. Is the space I^I separable, where $I = [0, 1]$?

9. State the Seifert–van Kampen Theorem.

10. State the Tychonoff Theorem.

11. Show that the projective plane $\mathbb{R}P^2$ is not homeomorphic to the surface of genus 2, $M_2 = T \# T$.

12. Let $X = [0, 1]^{(0,1)}$ be given the product topology. Is the space X

(a) compact?
(b) metrizable?
(c) separable?

13. Let $X = \mathbb{R}$ and $A = \mathbb{Z}$ be the reals and the integers respectively. Here X^* is the partition into A and singletons.

(a) Is the quotient space $X^* = X/A$ metrizable?
(b) What is $\pi_1(X^*)$?

14. Give definition of a deformation retraction.

15. If $r : X \to A$ is a retraction, show that the induced map for the fundamental groups $r_\#$ is surjective.